DAY 31 | POST-READ

Evaluating GenAI Systems: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
Evaluation loop	Dataset, run, score, analyse failures, improve one thing, repeat.
Automated vs human	Automation scales; humans label a sample and remain the reference.
Retrieval metrics	Hit rate at k, MRR, context relevance.
Generation metrics	Faithfulness (supported claims / all claims), answer relevance, correctness.
Hallucination types	Unsupported addition, contradiction, wrong source mixing, fabricated citation.
LLM as judge	Rubric, one criterion per call, reasons first; biases: position, verbosity, self-preference.
Judge calibration (NEW v4)	Human labels on a sample; agreement within 1 point and Cohen's kappa.
Trajectory evaluation (NEW v4)	Tool selection, order, extra and forbidden calls, steps, final answer.
pass^k (NEW v4)	Share of tasks solved on all k runs; pass@k is the share solved on at least one.
Security regression evals (NEW v4)	Block rate, false alarms and forbidden calls tracked per version.

Part B: Going beyond today

1. Ragas and other frameworks

Ragas packages common RAG metrics using LLM judges. Use it once you understand what each metric measures, as you now do. Check its documentation for the current API.

2. Evaluating on production traces

Offline sets catch regressions before release. After release, sample real traces, collect thumbs up and down, and add confusing real questions and failed trajectories to the eval set. Day 35 connects this to tracing.

3. Synthetic test generation

Models can draft questions from your documents to grow an eval set quickly. Always have people review them; synthetic sets tend to be easier and more literal than real user questions.

4. Inter-annotator agreement

Before trusting human labels, have two people label the same 30 items and compare. If humans disagree often, the rubric is unclear, and no judge model will fix that.

5. Multilingual quality

Embedding models and LLMs vary widely by language. If users write in Japanese or Hindi, include those languages in the eval set and slice every metric by language.

Part C: Check yourself

- Why evaluate retrieval and generation separately?
- How is faithfulness calculated?
- Why can an agent with a correct final answer still fail a trajectory evaluation?
- If one run succeeds 90% of the time, what is pass^3 roughly?
- How do you know an LLM judge can be trusted?

Part D: Practice ideas for tonight

- Add 10 real-style questions, including 3 in Japanese or Hindi, to the eval set.
- Label 15 answers yourself and compare with the offline or LLM judge.
- Write a short evaluation report: overall scores, slices, top three failure causes and fixes.

Coming up on Day 32: golden datasets, prompt regression testing, tool-call fidelity tests, and quality gates with token and cost metrics.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today